

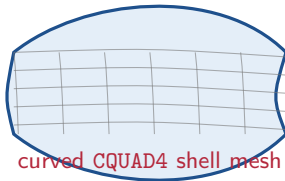
JFEM Shell-Element Formulation and Internal Gates

A Julia FEM solver with Nastran-compatible front end

Technical deck prepared from the checked-in source tree

JFEM 2026 workspace

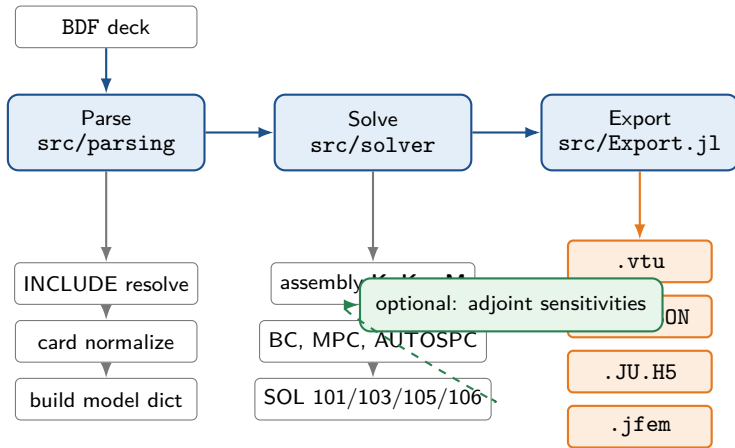
May 2026



Outline

- 1 JFEM at a glance — architecture
- 2 Solution sequences SOL 101/103/105/106
- 3 Shell kinematics, DOF layout, frames
- 4 Material handling: MAT1/2/8, PCOMP CLT
- 5 PSHELL field routing
- 6 Per-element CQUAD4 dispatcher
- 7 MITC4 default kernel
- 8 φ_2 shear anti-locking
- 9 MacNeal RBF, MITC4+, MITC4-3D
- 10 DKQ, DKMQ, Adini, MIN4, hybrid stress, Hu–Washizu
- 11 CTRIA3 macro-quad triangle
- 12 Higher-order CTRIA6/CQUAD8 subdivision
- 13 Geometric stiffness $\mathbf{K}_G$ for SOL 105
- 14 Stress recovery and consistent mass
- 15 Gate catalogue
- 16 Validation status and open items

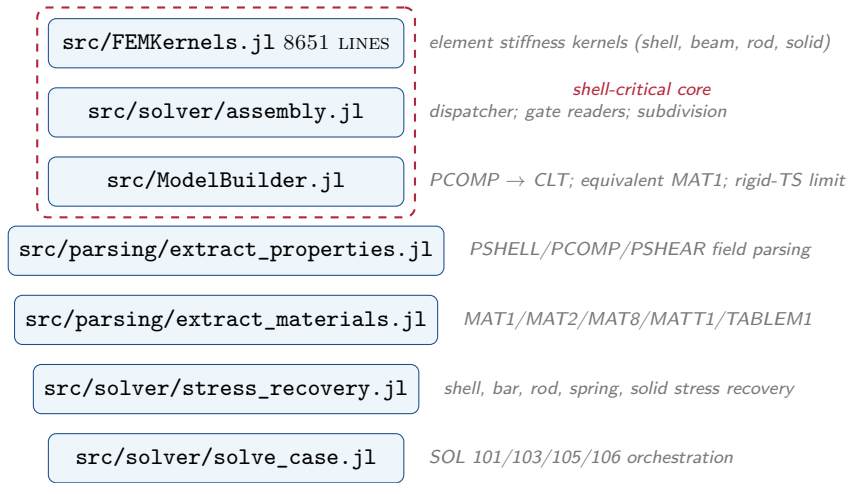
JFEM at a glance — three-stage workflow



Two design choices that drive everything

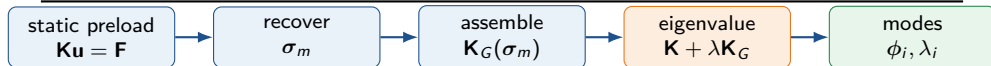
- The internal model is a JSON-serialisable nested Dict — the parse stage can be replaced by any external pre-processor.

Module map relevant to the shell layer



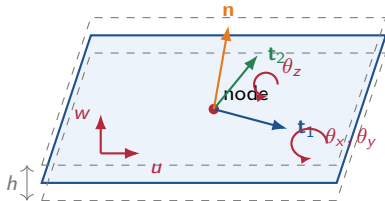
Solution sequences — equations and exports

SOL	Equation	Exports
101 static	$\mathbf{K}\mathbf{u} = \mathbf{F}$	displacements, reactions, $N/M/Q$, σ_{vM}
103 modal	$\mathbf{K}\phi = \omega^2\mathbf{M}\phi$	freqs, mode shapes, m_{eff}
105 buckling	$(\mathbf{K} + \lambda\mathbf{K}_G)\phi = \mathbf{0}$	buckling factors, mode shapes
106 nonlinear (exp.)	$\mathbf{K}_{\text{eff}}^{(k)}\Delta\mathbf{u} = -\mathbf{r}^{(k)}$	step history, residuals, line search



SOL 105 buckling path

Shell DOF layout (Reissner–Mindlin kinematics)



Displacement field:

$$\mathbf{u}(x, y, z) = \mathbf{u}_0(x, y) + z \boldsymbol{\theta}(x, y)$$

6 DOFs per node: $(u, v, w, \theta_x, \theta_y, \theta_z)$.

Transverse shear strains

$$\gamma_{xz} = \frac{\partial w}{\partial x} - \theta_y$$

$$\gamma_{yz} = \frac{\partial w}{\partial y} + \theta_x$$

Element sizes

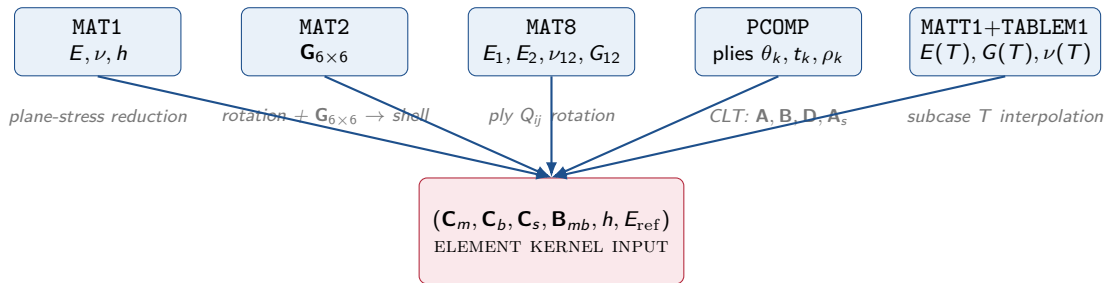
CQUAD4: 24×24

CTRIA3: 18×18

θ_z is penalised through

$$\alpha_{\text{drill}} = (\text{K6ROT}/10^5) Gh$$

Material hierarchy reaching the shell kernel



Special case — MAT8 with blank G_{1Z}/G_{2Z} in PCOMP:

Nastran convention $\Rightarrow$ rigid TS limit. JFEM default $\mathbf{C}_s = 2.5 \cdot \mathbf{C}_s^{\text{Whitney}}$.

Override: JFEM_PCOMP_RIGID_TS_CS_SCALE or disable with JFEM_PCOMP_RIGID_TS_DISABLE=true.

Isotropic _{MAT1} constitutive blocks

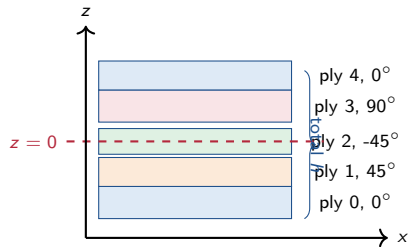
For PSHELL/MAT1 with (E, ν, h) and PSHELL fields $r_b=12I/T^3$ (default 1), $k_s=TS/T$ (default 5/6):

$$\mathbf{C}_m = \frac{Eh}{1-\nu^2} \begin{bmatrix} 1 & \nu & 0 \\ \nu & 1 & 0 \\ 0 & 0 & \frac{1-\nu}{2} \end{bmatrix}$$
$$\mathbf{C}_b = \frac{r_b Eh^3}{12(1-\nu^2)} \begin{bmatrix} 1 & \nu & 0 \\ \nu & 1 & 0 \\ 0 & 0 & \frac{1-\nu}{2} \end{bmatrix}$$
$$\mathbf{C}_s = k_s Gh \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad G = \frac{E}{2(1+\nu)}$$

Boundary cases

- MID2 blank $\Rightarrow \mathbf{C}_b = \mathbf{0}$, drill penalty also forced to 0 (*membrane-only shell*)
- MID1 blank $\Rightarrow \mathbf{C}_m = \mathbf{0}$ (*bending-only shell*)
- MID3 blank + MAT8 plies (PCOMP) $\Rightarrow$ rigid-TS override path

File evidence: FEMKernels.jl:1327-1335, extract_properties.jl:9-14.



Stack reduction (per laminate)

$$\mathbf{A}_{ij} = \sum_k \bar{\mathbf{Q}}_{ij}^{(k)} (z_k - z_{k-1})$$

$$\mathbf{B}_{ij} = \frac{1}{2} \sum_k \bar{\mathbf{Q}}_{ij}^{(k)} (z_k^2 - z_{k-1}^2)$$

$$\mathbf{D}_{ij} = \frac{1}{3} \sum_k \bar{\mathbf{Q}}_{ij}^{(k)} (z_k^3 - z_{k-1}^3)$$

$$\mathbf{A}_s = \sum_k \mathbf{G}_{sh}^{(k)} (z_k - z_{k-1})$$

Stored on the property as: $\mathbf{C}_m = \mathbf{A}$, $\mathbf{C}_b = \mathbf{D}$, $\mathbf{B}_{mb} = \mathbf{B}$ (or nothing if $\|\mathbf{B}\| < 10^{-10} \|\mathbf{A}\|$), $\mathbf{C}_s = k_s \mathbf{A}_s$.

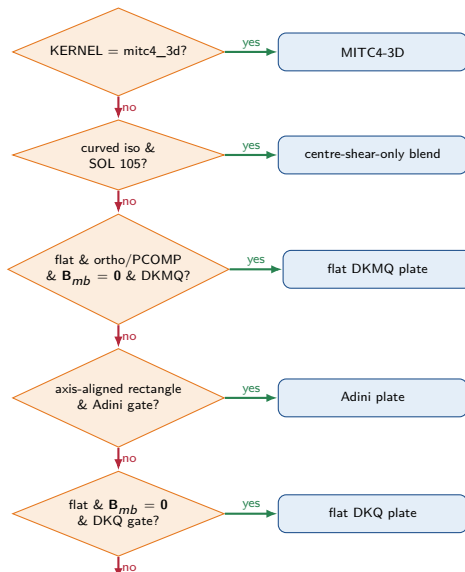
Synthesised side data

$E_{eq} = A_{11}(1 - \nu_{eq}^2)/h$, $\rho_{eff} = \sum_k \rho^{(k)} t^{(k)}/h$, PLY_DATA retained for per-ply stress recovery.

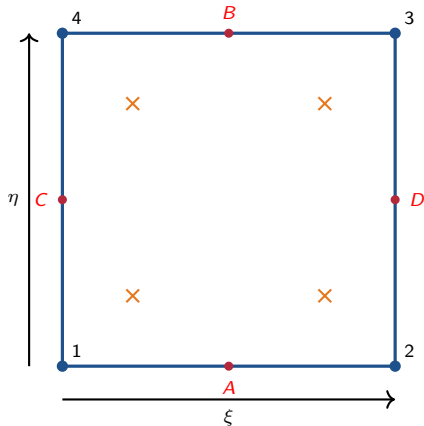
Field	Default	Role
MID1	—	membrane material id $\rightarrow \mathbf{C}_m$
T	—	thickness h (drives all thickness powers)
MID2	blank	bending material id; blank $\Rightarrow$ membrane-only
$12I/T^3$	1.0	bending ratio r_b on $\mathbf{C}_b$
MID3	blank	transverse-shear material; blank routes default
TS/T	5/6	transverse-shear factor k_s
NSM	0.0	non-structural mass per area $\rightarrow \mathbf{M}_{NSM}$

All field semantics are honoured directly from `extract_properties.jl:1-30`.

CQUAD4 dispatcher — decision tree



Default kernel: bilinear MITC4 with φ_2 anti-locking



- MITC4 tying pt
- 2x2 Gauss pt
- corner node

Membrane $\mathbf{B}_m$: bilinear $\partial N_a/\partial x, \partial N_a/\partial y$ at 2x2 Gauss, optional Wilson modes $\phi_1=1-\xi^2, \phi_2=1-\eta^2$.

Bending $\mathbf{B}_b$: θ_x, θ_y at 2x2 Gauss; optional MacNeal twist correction $2\mathbf{B}_b^{\text{GP}} - \mathbf{B}_b^c$.

Shear $\mathbf{B}_s$: tied at A, B, C, D, interpolated to Gauss; covariant strains rotated:

$$\mathbf{C}_s^{\text{phys}} = \varphi_2 (J^{-1})^\top \mathbf{C}_s J^{-1}$$

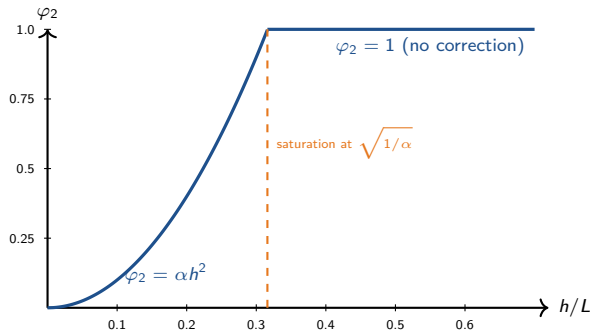
Anti-locking factor

$$\varphi_2(\xi, \eta) = \min\left(1, \alpha \frac{h^2}{4|\det J|}\right)$$

$\alpha = \text{PHI2_ALPHA}[]$ (default 10).

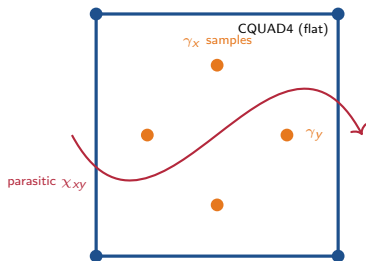
Drill: $\alpha_{\text{drill}} = \frac{k_6}{10^5} Gh$, $k_6 = \text{PARAM}, K6ROT$ (default 100).

φ_2 behaviour — thin vs. coarse meshes



- **Thin shell, fine mesh:** $h/L \ll 1 \Rightarrow \varphi_2$ drops, softening artificial shear stiffness $\rightarrow$ no shear locking.
- **Thick / coarse mesh:** $\varphi_2=1$, full Bathe–Dvorkin MITC4 is recovered.
- Saturation at $h/L = 1/\sqrt{\alpha} \approx 0.316$ with $\alpha=10$.
- Setting $\alpha=0$ disables the factor entirely (research mode).

Alternative: MacNeal residual-bending flexibility



Idea (MacNeal 1978): cancel parasitic-twist shear by augmenting $\mathbf{B}_s$ with a residual-bending flexibility block:

$$\tilde{\mathbf{B}}_s = \mathbf{B}_s^{\text{MITC4}} + \frac{L^2}{12EI} \mathbf{1}$$

implemented as an explicit 4×4 matrix $\mathbf{Z}_b$ inverted in closed form.

Default tuning (2026-05-14)

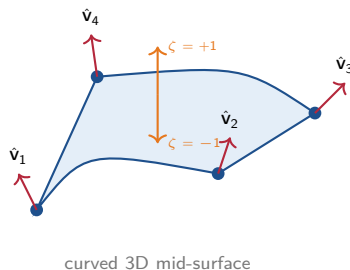
JFEM_Q4_MACNEAL_RBF_ZB_SCALE = 0.65

Legacy 1.0 restored via `..._ZB_SCALE_LEGACY=true`.

Eligibility: warp $< 10^{-4}$ (relative), no PCOMP curvature.

Bending-scale family of knobs (`_BENDING_SCALE`, `_ASPECT_SCALE`, `_CURVED_BENDING_SCALE`) calibrates per regime.

MITC4-3D — degenerate-shell kernel for curved geometry



Position field (Dvorkin–Bathe 1984):

$$\mathbf{x}(\xi, \eta, \zeta) = \sum_a N_a \left(\mathbf{x}_a + \frac{h}{2} \zeta \hat{\mathbf{v}}_a \right)$$

Directors $\hat{\mathbf{V}}_a$: from SNORM or geometric corner-normal average.

Operator entry points (FEMKernels.jl:658–900):

- `stiffness_quad4_mitc4_3d_resultant_matrices` — laminate-resultant path
- `stiffness_quad4_mitc4_3d_ply_matrices` — per-ply integration (PCOMP)

Selectors: `..._SHEAR_MODE`

(covariant/physical/macneal); per-block scales
`..._{MEMBRANE,BENDING,SHEAR,DRILL}_SCALE`.

Status: *opt-in*; planned promotion to default to close curved-PCOMP gap.

MITC4+ assumed natural-strain membrane (Ko–Lee–Bathe 2017)

Standard MITC4 carries a parasitic $\xi\eta$ membrane term on *distorted* elements (e.g. HTP_launch trapezoids). MITC4+ replaces the membrane field with an assumed natural-coordinate strain:

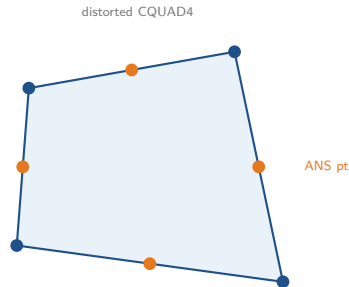
$$\varepsilon_{\alpha\beta}^{\text{ANS}} = \sum_{k=1}^4 N_k(\xi, \eta) \varepsilon_{\alpha\beta}^{(k)}$$

sampled at four ANS points, eliminating the locking mode.

Variants:

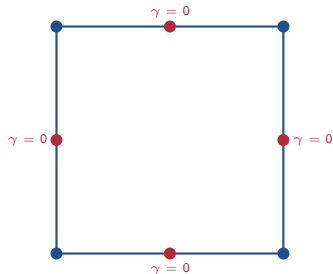
- :mitc4plus_linear_J — linear ANS using centre Jacobian
- :mitc4plus_quadratic_J — bilinear, J -dependent

Curved-shell variant gated by
JFEM_SOL105_EIG_MITC4PLUS_ALLOW_CURVED (default false).



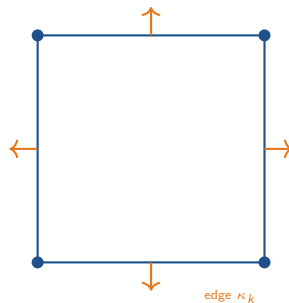
DKQ & DKMQ — discrete-Kirchhoff plate kernels

DKQ: $\gamma_{shear} = 0$ at edge midpoints



Thin-plate kernel. No transverse-shear deformability.
Active when flat & $\mathbf{B}_{mb} = \mathbf{0}$ & $\mathbf{C}_b \neq \mathbf{0}$.

DKMQ: per-edge hierarchical-rotation locking κ_k



$$\kappa_k = \frac{12}{L_k^2} \left[(\mathbf{H}_s^{-1})_{21} (\mathbf{H}_b)_{32} + (\mathbf{H}_s^{-1})_{22} (\mathbf{H}_b)_{22} \right]$$

Includes shear; orthotropic/PCOMP-aware bending–twist coupling.

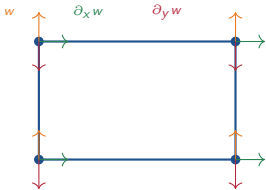
Default for flat orthotropic PCOMP.

Refs: Batoz/Bathe/Ho 1980, Katili–Maknun–Batoz–Ibrahimbegović 2018.

Adini rectangular plate & Tessler–Hughes MIN4

Adini (1961)

Axis-aligned rectangle only. 12-DOF bending: $(w, \partial_x w, \partial_y w)$ per corner. Full-cubic Hermite deflection, 4×4 Gauss–Legendre bending integration.



Fallback: non-rectangular $\rightarrow$ DKQ.

MIN4 (Tessler–Hughes 1983)

Anisoparametric: biquadratic w + bilinear (θ_x, θ_y) ; transverse-shear shape functions constrained to drop spurious mode.

Calibration knob

JFEM_MIN4_CBMIN4 = 3.6
(MYSTRAN reference)

Activation

JFEM_Q4_KERNEL = min4

Both kernels are **research switches**, available as opt-in alternatives to the MITC4 default.

Hybrid stress (Pian–Sumihara) & Hu–Washizu mixed

Hybrid stress (5 modes)

$$\mathbf{K}_e = \mathbf{G}^\top \mathbf{H}^{-1} \mathbf{G}$$
$$\mathbf{H} = \int P^\top S^{-1} P \, dA, \quad \mathbf{G} = \int P^\top B \, dA$$

Three variants in JFEM:

- ...membrane_hybrid_stress...
- ...bending_hybrid_stress...
- ...membrane_bending_hybrid_stress...

Hu–Washizu (3-field)

$$\Pi_{\text{HW}}(\mathbf{u}, \boldsymbol{\varepsilon}, \boldsymbol{\sigma}) \rightarrow \text{stationary}$$

Independent fields:

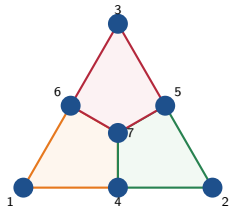
- displacement $\mathbf{u}$
- strain $\boldsymbol{\varepsilon}$ (5 modes)
- stress $\boldsymbol{\sigma}$ (5 modes)

Activation: JFEM_Q4_KERNEL=huwashizu.

Used as a *clean baseline* for cross-checking MITC4, MacNeal, MIN4.



CTRIA3 as a macro-quad superposition



Algorithm (FEMKernels.jl:6782-6862):

- 1 7 virtual points: 3 corners + 3 edge mids + 1 centroid
- 2 3 overlapping macro-quads:
TRIA3_MACRO_QUADS =
((1,4,7,6), (2,5,7,4), (3,6,7,5))
- 3 Each macro-quad runs through
stiffness_quad4_matrices
- 4 Keep plate DOFs only (w, θ_x, θ_y) per macro-quad
- 5 Assemble 21×21 macro stiffness
- 6 Condense the 12 internal DOFs $\rightarrow 9 \times 9$

$$\mathbf{K}_{\text{cond}} = \mathbf{K}_{ee} - \mathbf{K}_{ei} \mathbf{K}_{ii}^{-1} \mathbf{K}_{ie}$$

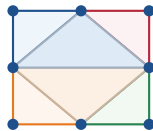
The condensation map $\mathbf{A}_{\text{int}} = -\mathbf{K}_{ii}^{-1} \mathbf{K}_{ie}$ is retained for pressure loads, stress recovery, and $\mathbf{K}_G$.

Higher-order shells via subdivision

CTRIA6 $\rightarrow$ 4 CTRIA3

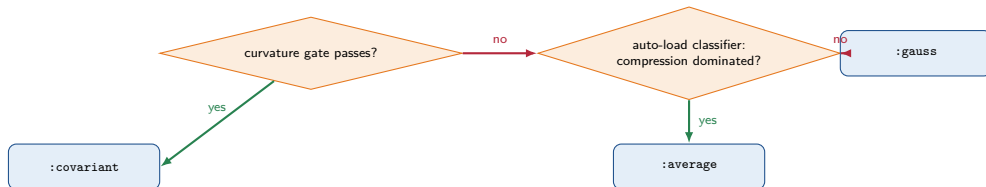


CQUAD8 $\rightarrow$ 6 CTRIA3



Quadratic basis *not exploited*; effective element order = 1. Trade-off: input compatibility at modest implementation cost. A dedicated quadratic kernel is a candidate development item, not scheduled.

K_G stress-field classifier (`JFEM_KG_QUAD4_STRESS_FIELD_MODE`)



Tests in the auto-load classifier (all must pass for averaged σ_m):

- per-element geometry test;
- both principal averages in compression;
- biaxial compression;
- shear-resultant ratio below 2%;
- global load dominated by one axis with the matching sign.

K_G extras: `rot_grad_scale`, principal-TS coupling (`JFEM_KG_SHELL_TRANS_MODE=principal`), `PCOMP` N_{xy} scale.

$\mathbf{K}_G$ assembly recipe for shells

Per element, per Gauss point:

$$\mathbf{K}_G^{(g)} = \int_A (\partial N)^\top \begin{bmatrix} \sigma_{xx} & \sigma_{xy} \\ \sigma_{xy} & \sigma_{yy} \end{bmatrix} (\partial N) dA$$

Per-mode contributions added on top:

translational
 $(\partial N)^\top \sigma (\partial N)$

rotational
rot-grad branch

principal TS
...TRANS_MODE

membrane-incomp
Wilson bubble

plate variants
(DKMQ, DKQ, Adini)

Stress recovery routes feeding $\mathbf{K}_G$

- quad4_membrane_force_field — 2×2 Gauss, optional Wilson condensation
- quad4_membrane_force_field_triangle_recovery — CST triangle averaging modes
- quad4_membrane_force_field_covariant — 3D coords + metric tensor
- quad4_membrane_cst_resultant — exact constant-strain triangle

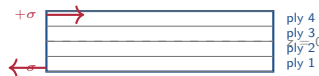
Shell stress recovery: what gets exported

Quantities recovered (`stress_recovery.jl:163–430`):

- membrane resultants (N_{xx} , N_{yy} , N_{xy})
- bending moments (M_{xx} , M_{yy} , M_{xy})
- transverse shear (Q_x , Q_y) at centre
- top/bottom surface σ_{xx} , σ_{yy} , σ_{xy} , σ_{VM}
- **Per-ply** stresses at top/bot of every ply (PCOMP), rotated back by θ_{ply}

Core helpers

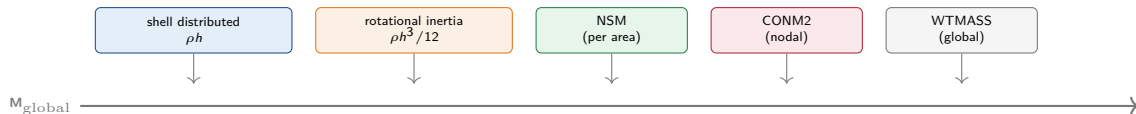
- `stress_strain_quad4` — full Q/T/B recovery
- `stress_strain_tria3` — via macro-quads + $\mathbf{A}_{int}$ map
- `quad4_mitc4_center_shear_resultant` — (Q_x , Q_y) at $\xi=\eta=0$



Resultant theory follows Simo–Fox–Rifai 1989:
 N , M , Q direct from constitutive blocks, not
through-thickness re-integration.

Consistent mass for shells

- `consistent_mass_quad4` (FEMKernels.jl:7857): 2×2 Gauss; translational $\propto \rho h$, rotational $\propto \rho h^3/12$, drilling = 0.
- `consistent_mass_tri3` (FEMKernels.jl:7903): analytical on the CST; diagonal $\rho h A/6$, off-diagonal $\rho h A/12$.
- Non-structural mass from PSHELL/NSM or PCOMP/NSM adds to diagonal as $\text{NSM} \cdot A$.
- `PARAM,WTMASS` multiplies the final **M**.
- `CONM2` concentrated mass added separately.



Internal gates — categories at a glance

Globals

PHI2_ALPHA
MITC4PLUS_ALLOW_CURVED

Curvature

...CURVED_JACOBIAN
...CURVATURE_FILTER

MacNeal RBF

...ZB_SCALE = 0.65
...BENDING_SCALE_*

MITC4-3D

...SHEAR_MODE
...{M,B,S,D}_SCALE

Flat-iso

...MEMBRANE_INCOMP
...DKMQ

Flat-PCOMP

...PLATE_BRANCH
...EXACT_MEMBRANE

Drill / K6ROT

PARAM,K6ROT=100
...EIG_FLOOR=300

PCOMP rigid-TS

...DISABLE
...CS_SCALE=2.5

Marguerre

...COUPLING
...CONVENTION

Kg

...STRESS_FIELD_MODE
...AUTO_AVG_LOAD_*

Static-component

...{CM,CB,CS}_SCALE
...EID_FILTER

Hotspot

...NEIGHBOR_PID
...V2_MIN/MAX

≈ 70 named env vars and PARAM entries, organised into 12 categories. Full catalogue in the companion paper (§12).

Example: switching default kernel & tuning rigid-TS

```
# Use MITC4-3D as the default kernel for SOL 105 eigenvalue assembly
export JFEM_Q4_KERNEL_EIG=mitc4_3d
export JFEM_Q4_MITC4_3D_PLY_INTEGRATION=true

# Override PCOMP rigid-TS uniform scale (default 2.5)
export JFEM_PCOMP_RIGID_TS_CS_SCALE=2.0

# Disable rigid-TS entirely (Whitney softening recovered)
export JFEM_PCOMP_RIGID_TS_DISABLE=true

# Localisation filter: scale Cs only on a named cluster of EIDs
export JFEM_Q4_STATIC_COMPONENT_EID_FILTER="51100,51101,51102"
export JFEM_Q4_STATIC_COMPONENT_CS_SCALE=2.5
export JFEM_Q4_STATIC_COMPONENT_NEIGHBOR_PID_PREFIXES="HTP_,VTP_"
```

Gates are read *once per element* at assembly time. Per-pass overrides (`..._STATIC`, `..._EIG`, `..._KG`) allow the static $\mathbf{K}$, eigen $\mathbf{K}$, and $\mathbf{K}_G$ to use different kernels in the same run.

Adjoint sensitivities — shell variables

Variable	Back-end	Interpretation
shell_thickness	:ad_forward	ForwardDiff on CTRIA3 + default CQUAD4
material_E	:analytical	Exact $\partial \mathbf{K} / \partial E$
material_NU	:ad_forward	ForwardDiff on standard shell; exact $\partial \mathbf{K}_G / \partial \nu$ on specialised branches
pcomp_ply_thickness	:laminate_exact	Differentiates $\mathbf{A}, \mathbf{B}, \mathbf{D}, \mathbf{A}_s$ directly
pcomp_ply_angle	:laminate_exact	Same path, with θ_{ply} AD
node_coord	:full_model_fd	Geometric perturbation + reassembly



Validation harness

Broad corpus (`tools/testing/run_mscval_compare.jl`):

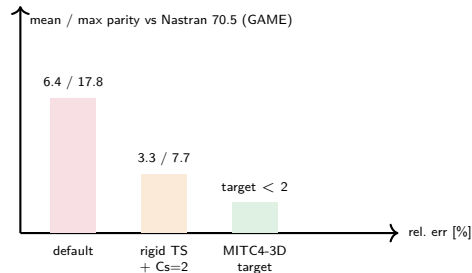
Category	Count	Interpretation
Total decks processed	3997	full Nastran F06 paired set
OK (workflow + parity)	2048	$\approx 67.8\%$ raw / 74.4% effective
Solver error	973	numerical/procedural failure
Unsupported sequence	976	outside current scope (SOL 200, etc.)
Effective fragment	267	standalone INCLUDE pieces

Focused regressions cover:

- shell-level $\partial \mathbf{K}_G / \partial h$, $\partial \mathbf{K}_G / \partial \nu$
- raw-JSON bridge parity (BDF $\rightarrow$ raw JSON $\rightarrow$ model $\rightarrow$ solve)
- per-element MAC-based curved-shell tracking
- SOL 106 readiness gate (`verify_sol106.jl`)

Open items relevant to shells

- **Curved-PCOMP coupling:** HTP_3wp_strain MAC= 0 on subcase 511002. Membrane-recovery resultant ratio 0.4 on strong-curvature patches.
- **Planned remedy:** promote MITC4-3D to default (acceptance criteria in MITC4_3D_SHELL_IMPLEMENTATION_PLAN.md).
- **Higher-order shells:** CTRIA6/CQUAD8 sub-divided, not natively quadratic.
- **SOL 106:** limited to flat coplanar PSHELL CTRIA3/CQUAD4.
- **Triangle drill:** inherits from macro-quad, no independent triangular formulation.



Key takeaways

- 1 JFEM is a **three-stage Julia FE solver** with Nastran-compatible BDF input and JSON-serialisable internal model.
- 2 Shell layer is **kernel-rich**: MITC4 (default) + MITC4+ + MITC4-3D + MacNeal RBF + DKQ + DKMQ + Adini + MIN4 + hybrid-stress + Hu-Washizu.
- 3 All shell properties — MAT1, MAT2, MAT8, PCOMP — reach the kernel as a uniform $(\mathbf{C}_m, \mathbf{C}_b, \mathbf{C}_s, \mathbf{B}_{mb})$ tuple plus per-ply data.
- 4 Kernel choice is a **per-element dispatcher decision**; ≈ 70 env vars + PARAMs tune it; defaults are conservative and validated.
- 5 SOL 105 $\mathbf{K}_G$ has a **stress-field classifier** that picks Gauss / averaged / covariant recovery per element.
- 6 **CTRIA3** is built as a 3-macro-quad superposition with static condensation, sharing the entire CQUAD4 kernel ladder.
- 7 Adjoint sensitivities: **mixed analytical + AD + FD**, with exact laminate-level paths for PCOMP ply variables.
- 8 Next big step: **MITC4-3D as default** for curved PCOMP, closing the largest remaining parity gap against Nastran.

Selected references

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Thank you

Questions?

